Python Programming Lab. [Batch-B3]

Assignment Submission Details

Field	Details
Student Name *	Dilip Balu Magar
Student PRN *	125M1H055
Course Name	Python Programming Lab. (MCA31PC06)
Academic Year	2025-26 (Semester-1)
Course Teacher	Prof. Prakash Ukhalkar
Assignment Name	Assignments based on modules in Python 1.
Assignment Number	Assignment 06
Submission Date *	17-11-2025

1.Write a Python program to list all files and directories in the current working directory using the os module

In [2]:
```
import os

print(f"Showing data of: {os.getcwd()}")

data = os.listdir('.')

files = []       # to store files
directories = [] # to store directories

for entry in data:

    if os.path.isdir(entry):
        directories.append(entry)

    elif os.path.isfile(entry):
        files.append(entry)


print("\nDirectories Found:")
for d in directories:
    print(f"{d}")

print("\nFiles Found:")
for f in files:
    print(f"{f}")
```

```
Showing data of: c:\Users\dilip\OneDrive\Desktop\Desktop\FYMCA\PYTHON

Directories Found:
.dist
learn-python-from-scratch

Files Found:
125M1H055_A02_PPLAB_B3.html
125M1H055_A02_PPLAB_B3.ipynb
125M1H055_A04_PPLAB_B3.html
125M1H055_A04_PPLAB_B3.ipynb
125M1H055_A04_PPLAB_B3.pdf
125M1H055_A05_PP_LAB_B3.ipynb
125M1H055_A06_PP_LAB_B3.ipynb
125M1H055_A07_PP_LAB_B3.ipynb
assignments.txt
basics.py
q1.ipynb
q2.ipynb
q3.ipynb
```

2 Create a Python script that accepts command-line arguments and prints the total number of arguments using the sys module

In [ ]:
```
import sys

argument_list = sys.argv

total_elements = len(argument_list)

print(f"given rgument list : {argument_list}")
print(f"total no. of arguaments: {total_elements}")
print(f"no.of arguments provided by user: {total_elements-1}")
#ouput is incorrect as this is .ipynb the code is work for .py file
```

```
given rgument list : ['C:\\Users\\dilip\\AppData\\Roaming\\Python\\Python312\\site-p
ackages\\ipykernel_launcher.py', '--f=c:\\Users\\dilip\\AppData\\Roaming\\jupyter\\r
untime\\kernel-v3d7ae00e1f57de34f189748c6d150a2887000eeff.json']
total no. of arguaments: 2
no.of arguments provided by user: 1
```

3.Write a Python program that creates a directory, changes the current working directory to it, and then removes it using the os module.

In [8]:
```
import os  # for getcwd(),mkdir(),chdir(),rmdir()


DIR_NAME = "new_folder"

original_dir = os.getcwd()

print(f"Current working directory: {original_dir}")


print(f"\ncreating new directory with name {DIR_NAME}")
```

```
os.mkdir(DIR_NAME)
print(f"\ndirectory with name {DIR_NAME} is created")

print(f"changing CWD to {DIR_NAME}")
os.chdir(DIR_NAME)
print(f"New Current working directory: {os.getcwd()}")


print("\nChanging CWD back to previous original directory...")
os.chdir(original_dir)
print(f"Now CWD : {os.getcwd()}")


print(f"\nRemoving directory: {DIR_NAME}")
os.rmdir(DIR_NAME)
print("Directory removed")
```

```
Current working directory: c:\Users\dilip\OneDrive\Desktop\Desktop\FYMCA\PYTHON

creating new directory with name new_folder

directory with name new_folder is created
changing CWD to new_folder
New Current working directory: c:\Users\dilip\OneDrive\Desktop\Desktop\FYMCA\PYTHON
\new_folder

Changing CWD back to previous original directory...
Now CWD : c:\Users\dilip\OneDrive\Desktop\Desktop\FYMCA\PYTHON

Removing directory: new_folder
Directory removed
```

4.Develop a script to fetch and display the current system path (sys.path) and environment variables using the osmodule.Write a Python program to find the size of a file in bytes using the os module.

In [29]:

```
import os
import sys


print("\nSystem Path:")
print(sys.path)
print("*"*70)



print("\nALL Environment Variables:")
i=1
for key, value in os.environ.items():
    print(f"  {key}: {value}")
    i=i+1
    if i>10:
        break

print("*"*70)
```

```
file_name = "assignments.txt"
file_size = os.path.getsize(file_name)

print(f"  File: {file_name}")
print(f"  Size: {file_size} bytes")
```

```
System Path:
['c:\\Program Files\\Python312\\python312.zip', 'c:\\Program Files\\Python312\\DLL
s', 'c:\\Program Files\\Python312\\Lib', 'c:\\Program Files\\Python312', '', 'C:\\Us
ers\\dilip\\AppData\\Roaming\\Python\\Python312\\site-packages', 'C:\\Users\\dilip
\\AppData\\Roaming\\Python\\Python312\\site-packages\\win32', 'C:\\Users\\dilip\\App
Data\\Roaming\\Python\\Python312\\site-packages\\win32\\lib', 'C:\\Users\\dilip\\App
Data\\Roaming\\Python\\Python312\\site-packages\\Pythonwin', 'c:\\Program Files\\Pyt
hon312\\Lib\\site-packages']
**********************************************************************

ALL Environment Variables:
  ALLUSERSPROFILE: C:\ProgramData
  APPDATA: C:\Users\dilip\AppData\Roaming
  APPLICATION_INSIGHTS_NO_STATSBEAT: true
  ASL.LOG: Destination=file
  CHROME_CRASHPAD_PIPE_NAME: \\.\pipe\crashpad_31428_WTRZZCAHTJSDUPMX
  COMMONPROGRAMFILES: C:\Program Files\Common Files
  COMMONPROGRAMFILES(X86): C:\Program Files (x86)\Common Files
  COMMONPROGRAMW6432: C:\Program Files\Common Files
  COMPUTERNAME: DILIP
  COMSPEC: C:\WINDOWS\system32\cmd.exe
**********************************************************************
  File: assignments.txt
  Size: 1127 bytes
```

5.Write a Python program to calculate the area of a circle using the math module.

In [16]:
```
import math  # for value of pi


radius = float(input("Enter radius of the circle: "))


area = math.pi * (radius ** 2)

print(f"Radius: {radius}")
print(f"Area of circle: {area:.4f} ")
```

```
Radius: 5.5
Area of circle: 95.0332
```

6.Implement a Python function to compute the factorial of a number using the math module.

In [18]:
```
import math  #for factorial()


number = int(input("Enter a number: "))
```

```
result = math.factorial(number)


print(f"\nFactorial of {number} is: {result}")
```

```
Factorial of 5 is: 120
```

7.Write a Python program to generate a list of prime numbers up to a given number using the math.isqrt() method

In [21]:
```
import math

n=int(input("enter a number"))
print(f"Prime Numbers up to {n} ")

primes = []


def is_prime(num):
    if num <= 1:
        return False

    max_divisor = math.isqrt(num)

    for i in range(2, max_divisor + 1):
        if num % i == 0:
            return False
    return True

for number in range(2, n + 1):
    if is_prime(number):
        primes.append(number)

print(primes)
```

```
Prime Numbers up to 50
[2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47]
```

8. Create a Python function that calculates the square root of a number and rounds it to two decimal places using the math module.

In [22]:
```
import math

number = float(input("Enter a non-negative number: "))


sqrt = math.sqrt(number)


result = round(sqrt, 2)

print(f"\nOriginal Number: {number}")
```

```
print(f"Square Root: {sqrt}")
print(f"Result after rounding square root to two dedimal places: {result}")
```

```
Original Number: 35.0
Square Root: 5.916079783099616
Result after rounding square root to two dedimal places: 5.92
```

9.Write a Python program to compute the greatest common divisor (GCD) of two numbers using math.gcd()

In [24]:
```
import math

num1 = int(input("Enter the first number: "))
num2 = int(input("Enter the second number: "))

result = math.gcd(num1, num2)

print(f"\nThe numbers are {num1} and {num2}.")
print(f"The GCD is: {result}")
```

```
The numbers are 15 and 12.
The GCD is: 3
```

10. Write a Python program to print the current date and time using the datetime module.

In [28]:
```
import datetime

current_time = datetime.datetime.now()


print(f"Current Date: {current_time.date()}")
print(f"Current Time: {current_time.strftime('%Y-%m-%d %H:%M:%S')}")
```

```
Current Date: 2025-11-17
Current Time: 2025-11-17 23:47:45
```